

Module 1 — Synthesis Quiz

Question bank

Module 1, Synthesis quiz

Readings: [Read 0](#), [Read 1](#), [Read 2](#)

Labs: [Demo](#), [Exercise](#)

Section A — Hypotheses and BRP (Read 2 §1)

A1. ESP hypotheses

ESP test: 25 trials, 5 symbols, guess each trial. Let p = probability of a correct guess on one trial.

1. Write H_0 and H_a in symbols.
2. In words: if H_0 is true? If we believe H_a ?

A2. ESP BRP conditions

ESP test: 25 trials, 5 symbols, one guess per trial. Assume trials are independent under H_0 (open deck / replacement) and random guessing gives $p = 0.2$ per trial.

List the **four BRP conditions** and briefly say why each holds in this setting.

Answer key — Section A — Hypotheses and BRP (Read 2 §1)

1. $H_0 : p = 0.2$ vs $H_a : p > 0.2$. H_0 : guessing only (no ESP). H_a : better than chance.
 2. Binary (correct/incorrect); fixed $n = 25$; fixed $p = 0.2$ under null; independent trials (open deck).
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Section B — n , p , and tail direction (Read 2 §2)

B1. n and p under H_0

Scenario: An online ESP test uses **5 symbols** per trial and **25 trials**. Under random guessing, $p = 0.2$ on each trial.

Under H_0 , what are n and p for the binomial count $X = \text{number correct}$?

B2. Observed data and p-value direction

ESP test: 25 trials, 5 symbols, random guessing under H_0 so $p = 0.2$. Test $H_0 : p = 0.2$ vs $H_a : p > 0.2$. You observe **7** correct out of 25.

1. What is x ?
2. Write the p-value as $P(X \geq \dots)$ or $P(X \leq \dots)$.
3. Which tail(s) of the null distribution?

Answer key — Section B — n , p , and tail direction (Read 2 §2)

1. $n = 25$; $p = 0.2$.
2. $x = 7$; $P(X \geq 7)$; **right tail**.

Section C — Null plots: mark data and shade p-value (Read 2 §3)

```
if (!requireNamespace("ggplot2", quietly = TRUE)) {  
  install.packages("ggplot2", repos = "https://cloud.r-project.org")  
}  
library(ggplot2)
```

Each plot below shows the **null distribution** (x vs $P(X = x)$ under H_0). Plots are **unshaded** — on your copy (print or sketch), add a **vertical line** at the observed x and **shade the p-value region** in the direction of H_a .

Run the R chunks to reproduce each null distribution in Colab or RStudio (same pattern as [Project 1](#) and the [demo lab](#)).

C1. Left-handed students

Research question: Is the proportion of left-handed students at our school higher than the national rate of 10%?

Parameter: p = proportion of students at our school who are left-handed.

Hypotheses: $H_0 : p = 0.10$ vs $H_a : p > 0.10$. Under H_0 , use $n = 25$ and the null value of p shown in the R chunk.

Data: observed $x = 6$.

```
n <- 25  
p <- 0.10  
null_df <- data.frame(x = 0:n, prob = dbinom(0:n, size = n, prob = p))  
ggplot(null_df, aes(x = x, y = prob)) +  
  geom_col() +  
  labs(x = "x", y = expression(P(X == x))) +  
  theme_minimal()
```

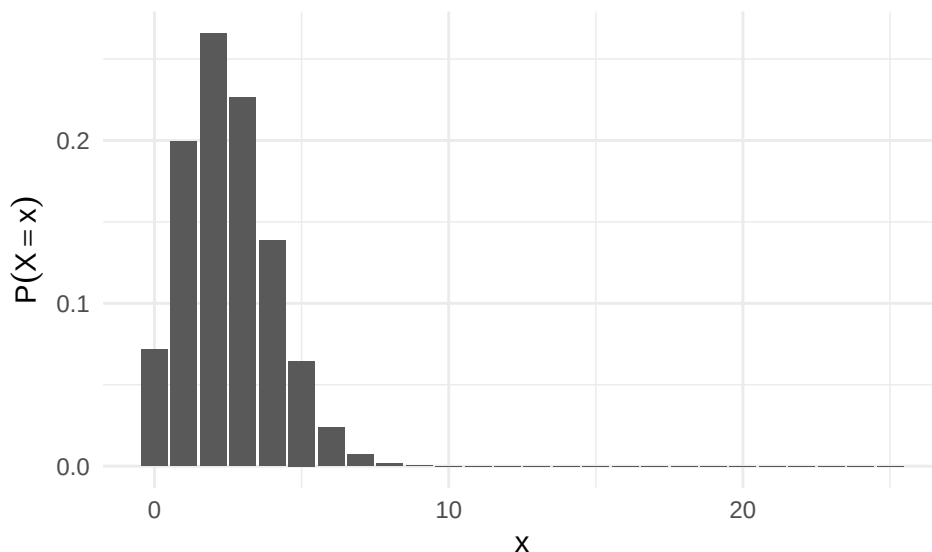


Figure 1: Null distribution: left-handed students, $n = 25$, $p = 0.10$ under H_0

1. On the plot (or a printout), add a **vertical line** at $x = 6$ and **shade** the p-value region (right tail).
2. p-value = $P(X \geq 6 \mid H_0)$ is **small** or **large**?
3. Conclusion in context (one sentence).

C2. Defective items

Research question: Is the proportion of defective items from the new supplier less than 15%?

Parameter: p = proportion of items from the new supplier that are defective.

Hypotheses: $H_0 : p = 0.15$ vs $H_a : p < 0.15$. Under H_0 , use $n = 25$ and the null value of p shown in the R chunk.

Data: observed $x = 3$.

```
n <- 25
p <- 0.15
null_df <- data.frame(x = 0:n, prob = dbinom(0:n, size = n, prob = p))
ggplot(null_df, aes(x = x, y = prob)) +
  geom_col() +
  labs(x = "x", y = expression(P(X == x))) +
  theme_minimal()
```

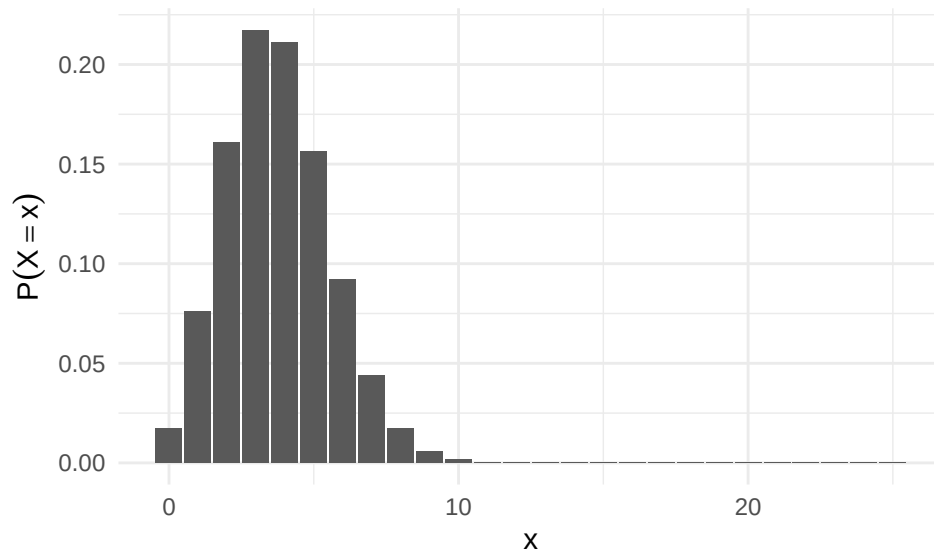


Figure 2: Null distribution: defective items, $n = 25$, $p = 0.15$ under H_0

1. On the plot (or a printout), add a **vertical line** at $x = 3$ and **shade** the p-value region (left tail).
2. p-value = $P(X \leq 3 \mid H_0)$ is **small** or **large**?
3. Conclusion in context (one sentence).

C3. Dice (sum 7 or 11)

Research question: When we roll two dice, is the probability that the sum is 7 or 11 equal to the fair-dice value ($8/36$)?

Parameter: p = probability that one roll of two dice gives sum 7 or 11.

Hypotheses: $H_0 : p = 8/36$ vs $H_a : p \neq 8/36$. Under H_0 , use $n = 25$ and the null value of p shown in the R chunk.

Data: observed $x = 2$.

```
n <- 25
p <- 8/36
```

```

null_df <- data.frame(x = 0:n, prob = dbinom(0:n, size = n, prob = p))
ggplot(null_df, aes(x = x, y = prob)) +
  geom_col() +
  labs(x = "x", y = expression(P(X == x))) +
  theme_minimal()

```

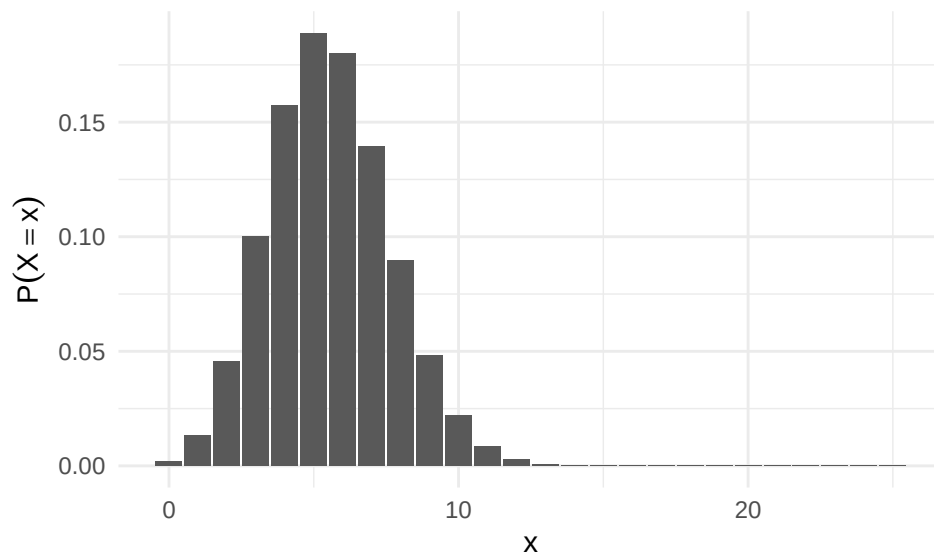


Figure 3: Null distribution: dice sum 7 or 11, $n = 25$, $p = 8/36$ under H_0

1. On the plot (or a printout), add a **vertical line** at $x = 2$ and **shade** the p-value region (both tails (two-sided)).
2. p-value = $P(X \leq 2) + P(X \geq 9)$ under H_0 is **small** or **large**?
3. Conclusion in context (one sentence).

C4. Gym use

Research question: Do more than 20% of students at our school use the gym at least once per week?

Parameter: p = proportion of students who use the gym at least once per week.

Hypotheses: $H_0 : p = 0.20$ vs $H_a : p > 0.20$. Under H_0 , use $n = 25$ and the null value of p shown in the R chunk.

Data: observed $x = 5$.

```

n <- 25
p <- 0.20
null_df <- data.frame(x = 0:n, prob = dbinom(0:n, size = n, prob = p))
ggplot(null_df, aes(x = x, y = prob)) +
  geom_col() +
  labs(x = "x", y = expression(P(X == x))) +
  theme_minimal()

```

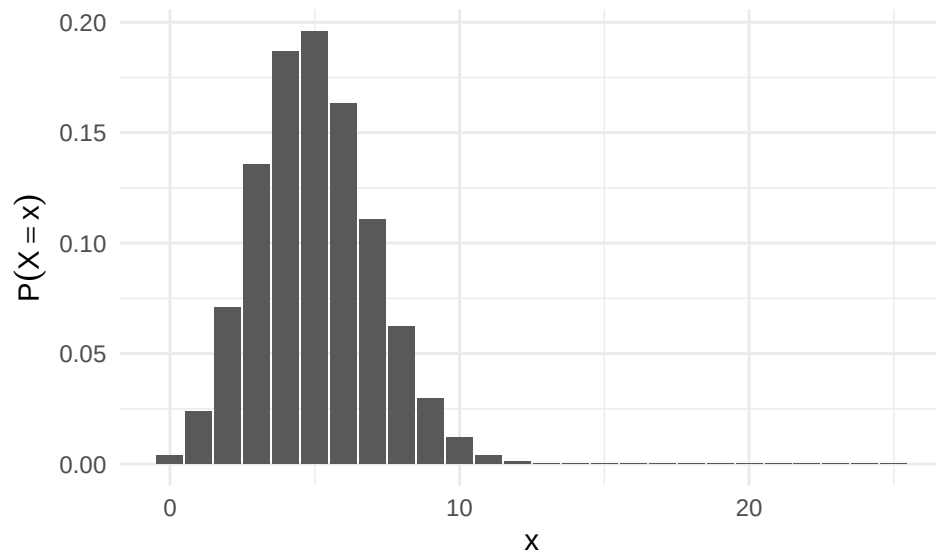


Figure 4: Null distribution: gym use, $n = 25$, $p = 0.20$ under H_0

1. On the plot (or a printout), add a **vertical line** at $x = 5$ and **shade** the p-value region (right tail).
2. p-value = $P(X \geq 5 \mid H_0)$ is **small** or **large**?
3. Conclusion in context (one sentence).

Answer key — Section C — Null plots: mark data and shade p-value (Read 2 §3)

1. Shade bars $x \geq 6$; **small** (about 0.04); reject H_0 — evidence proportion at school **exceeds** 10%.
 2. Shade $x \leq 3$; **large** (about 0.5); **fail to reject** — data consistent with 15% defect rate.
 3. Shade $x \leq 2$ and $x \geq 9$; typically **large**; **fail to reject** fair-dice proportion.
 4. Shade $x \geq 5$; **large** (about 0.5); **fail to reject** — no strong evidence share exceeds 20%.
-

Section D — Test workflow and one- vs two-sided (Read 2 §3)

D1. Order the steps

Number these binomial-test steps **1–4** in order:

- **(A)** Verify the four BRP conditions.
- **(B)** State research question, parameter p , H_0 , H_a , and n , p under H_0 .
- **(C)** Plot null distribution (x vs $P(X = x)$ under H_0).
- **(D)** Collect x , add observed value to plot, shade p-value region, compute p-value, conclude.

D2. One- vs two-sided shading

1. One-sided H_a (e.g. $p > 0.2$): shade **one tail** or **both tails**?
2. Two-sided H_a (e.g. $p \neq 8/36$): shade **one tail** or **both tails**?

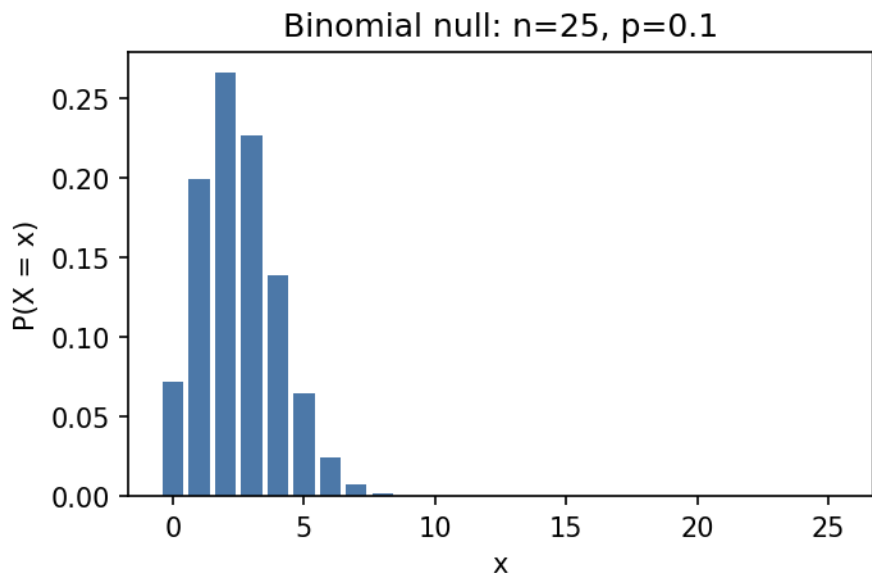
Answer key — Section D — Test workflow and one- vs two-sided (Read 2 §3)

1. 1=B, 2=A, 3=C, 4=D.

2. 1. **One tail** (direction of H_a). 2. **Both tails**.

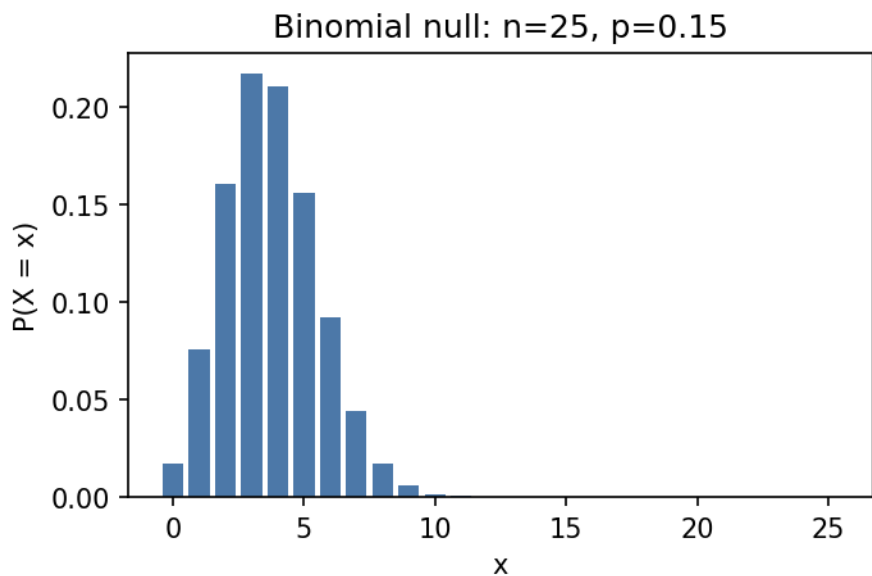
Section D2 — Binomial null plots (image interpretation)

IMG1. Binomial right-tail p-value region



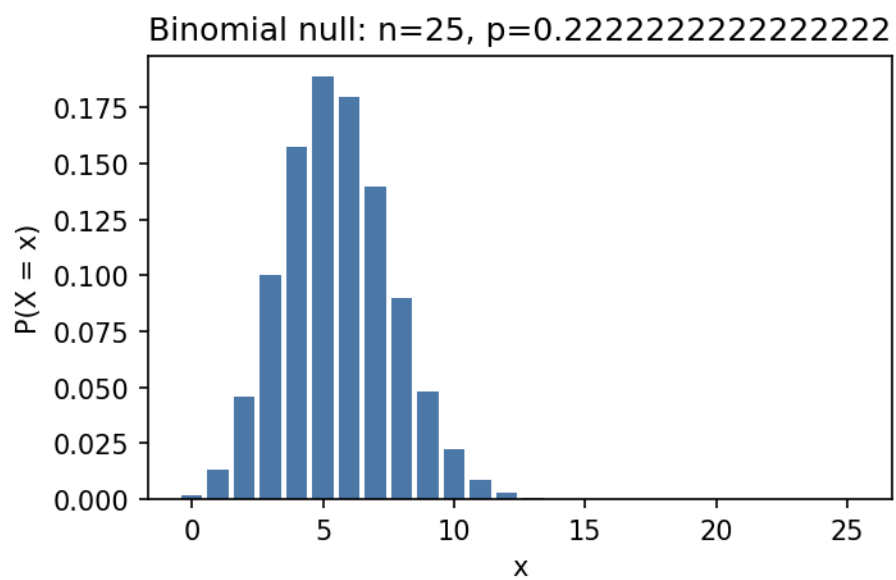
Observed $x = 6$, and $H_a : p > p_0$. Which bars are in the p-value region?

IMG2. Binomial left-tail p-value region



Observed $x = 2$, and $H_a : p < p_0$. Which bars are in the p-value region?

IMG3. Binomial two-sided p-value region



Observed $x = 2$, and $H_a : p \neq p_0$. Which region is the p-value?

Answer key — Section D2 — Binomial null plots (image interpretation)

1. Bars at and to the **right** of $x = 6$ (right tail for $H_a : p > p_0$).
 2. Bars at and to the **left** of $x = 2$ (left tail for $H_a : p < p_0$).
 3. **Both tails**: values as or more extreme than $x = 2$ in each direction.
-

Section E — Binomial R and ggplot (after demo lab)

E1. dbinom

One line: $P(X = 5)$ when $X \sim \text{Binomial}(10, 0.4)$. Store in `prob_five`.

E2. pbinom lower tail

One line: $P(X \leq 3)$ for the same distribution. Store in `prob_at_most_3`.

E3. pbinom upper tail

One line: $P(X > 2)$ using `lower.tail = FALSE`. Store in `prob_more_than_2`.

E4. rbinom

One line: simulate 1000 values from $\text{Binomial}(8, 0.2)$. Store in `sims`.

E5. Estimate from simulation

Using `sims` from E4, one line to estimate $P(X \leq 2)$. Store in `p_est`.

E6. aes() in ggplot2

What does `aes()` do? Choose the best answer:

- a) Creates the axes.
- b) Maps variables to visual properties (x, y, color, fill).
- c) Chooses the geometry (`geom_col`, etc.).
- d) Defines the data frame.

E7. Null bar plot with geom_col

Fill in blanks using: `0:10 | dbinom(0:10, size = 10, prob = 0.4) | x | prob`

```
df <- data.frame(x = _____, prob = _____)
ggplot(df, aes(x = _____, y = _____)) + geom_col()
```

Answer key — Section E — Binomial R and ggplot (after demo lab)

1. `prob_five <- dbinom(5, size = 10, prob = 0.4)`
 2. `prob_at_most_3 <- pbinom(3, size = 10, prob = 0.4)`
 3. `prob_more_than_2 <- pbinom(2, size = 10, prob = 0.4, lower.tail = FALSE)`
 4. `sims <- rbinom(1000, size = 8, prob = 0.2)`
 5. `p_est <- mean(sims <= 2)`
 6. (b) — maps variables to visual properties.
 7. `df <- data.frame(x = 0:10, prob = dbinom(0:10, size = 10, prob = 0.4))` then `aes(x = x, y = prob)`.
-

Section E2 — dbinom in word problems (after demo lab)

J1. Free throws — what does dbinom compute?

Scenario: A basketball player takes **20 free throws**. Under a practice model $p = 0.75$ (75% make rate on one shot), let X = number of makes.

In R: `dbinom(16, size = 20, prob = 0.75)`

1. What is n ? p ? The x in the first argument?
2. What probability does this line compute (in words)?

J2. Defect rate — write dbinom

Scenario: A supplier claims **15%** of items are defective. You inspect **25 items** and want $P(X = 3)$ defectives if $X \sim \text{Binomial}(25, 0.15)$.

Write one R line (you may use `size` and `prob` names).

J3. ESP — null bar height

Scenario: ESP test: **25 trials**, **5 symbols**, null $H_0 : p = 0.20$. You plot the null distribution; bar height at $x = 5$ is $P(X = 5)$ under H_0 .

Write the `dbinom` expression for that bar height (do not plot).

J4. Match arguments to the story

Scenario: **30 students** surveyed; under H_0 each has probability **0.20** of weekly gym use. You evaluate `dbinom(9, size = 30, prob = 0.20)`.

Match each R argument to the story:

- First argument 9 $\rightarrow$?
- `size = 30` $\rightarrow$?
- `prob = 0.20` $\rightarrow$?

Answer key — Section E2 — dbinom in word problems (after demo lab)

1. $n = 20$, $p = 0.75$, $x = 16$. Computes $P(X = 16)$ — exactly 16 makes in 20 shots.
 2. `dbinom(3, size = 25, prob = 0.15)`
 3. `dbinom(5, size = 25, prob = 0.2)`
 4. $9 = \text{count } x$; `size = 30 = n` trials; `prob = 0.20` = null p under H_0 .
-

Section E3 — pbinom in word problems (after demo lab)

K1. At most 3 defectives

Scenario: Defect inspection: **25 items**, null $p = 0.15$, X = number defective. You want $P(X \leq 3)$.

Write one R line using `pbinom`.

K2. At least 7 correct guesses

Scenario: ESP: **25 trials**, $H_0 : p = 0.20$, $H_a : p > 0.20$. Observed **7** correct. The p-value uses the right tail $P(X \geq 7)$.

Write one R line using `pbinom` and `lower.tail = FALSE`.

K3. What does this pbinom line mean?

Scenario: Coin flipped **20 times**; null $p = 0.50$ for heads; X = number of heads.

`pbinom(14, size = 20, prob = 0.5, lower.tail = FALSE)`

Write the probability statement in symbols ($P(X \dots)$).

Answer key — Section E3 — pbinom in word problems (after demo lab)

1. `pbinom(3, size = 25, prob = 0.15)`
 2. `pbinom(6, size = 25, prob = 0.2, lower.tail = FALSE)` — $P(X \geq 7) = P(X > 6)$.
 3. $P(X \geq 15)$ or $P(X > 14)$ — upper tail starting at 15 heads.
-

Section F — R basics (Module 0 demo / exercise)

F1. Arithmetic

One line of R that produces **10** by adding two numbers.

F2. Multiplication

One line of R that produces **6** by multiplying two numbers.

F3. Assignment and `sqrt`

Assign **16** to `x`, then compute $\sqrt{x}$.

F4. Vector

Create vector `numbers` with values 2, 4, 6, 8, 10.

F5. Indexing

Using `numbers`, return the **third** element.

F6. Vector arithmetic

Create `numbers_plus_one` by adding 1 to every element of `numbers`.

F7. Mean

Compute the mean of `numbers`.

F8. Count with `==` and `sum`

Given `flips <- c("H","T","H","H","T","H")` and `heads <- c("H","H","H","H","H","H")`, count how many "H" appear using `==` and `sum()`.

Answer key — Section F — R basics (Module 0 demo / exercise)

1. Any valid addition, e.g. `3 + 7` or `5 + 5`.
 2. e.g. `2 * 3`.
 3. `x <- 16` then `sqrt(x)` (two lines or one).
 4. `numbers <- c(2, 4, 6, 8, 10)`
 5. `numbers[3]`
 6. `numbers_plus_one <- numbers + 1`
 7. `mean(numbers)`
 8. `sum(flips == heads)` or `sum(flips == "H")` → 4.
-

Section G — Vectors, matrices, loops (demo Section B / exercise Part I)

G1. Types and `class()`

Create `x_num <- 0`, `x_int <- 0L`, and print the class of each.

G2. Vector and `length`

Create `v <- c(2,4,6,8,10)`, print `length(v)`, create `v_plus_one <- v + 1`.

G3. Bracketing

With `v <- c(2,4,6,8,10)`, write code for: (1) 3rd element; (2) elements 2–4; (3) all except 3rd; (4) elements > 5 .

G4. Matrix and `dim()`

Create 2×3 matrix `m` from `1:6` (by column), print `dim(m)`, extract row 2 col 1, first row, second column.

G5. `for` loop squares

Use a `for` loop so `squares` contains $1^2, 2^2, \dots, 5^2$.

G6. Simulate $P(\text{even die})$

Estimate $P(\text{even die})$ with `n <- 10000`, `rolls <- sample(1:6, size=n, replace=TRUE)`, and `mean(...)`. Name objects `n`, `rolls`, `p_even`.

Answer key — Section G — Vectors, matrices, loops (demo Section B / exercise Part I)

1. `x_num <- 0; x_int <- 0L; class(x_num); class(x_int).`
 2. As stated; `length(v)` returns 5.
 3. `v[3]; v[2:4]; v[-3]; v[v > 5].`
 4. `m <- matrix(1:6, nrow=2, ncol=3); dim(m); m[2,1]; m[1,]; m[,2].`
 5. `squares <- numeric(5); for (i in 1:5) squares[i] <- i^2` (or `squares <- i^2` inside loop).
 6. `p_even <- mean(rolls %% 2 == 0)` or `mean(rolls %in% c(2,4,6)).`
-

Section H — Birthday matching simulation (demo / exercise)

H1. `rep()` for balls

Create vector `balls`: 5 red and 2 white, labels "R" and "W".

H2. Simulate without replacement

Bag: 3 red, 3 white. Simulate 10,000 draws of 2 without replacement; success = white first, red second; estimate with `mean(successes)`.

H3. `which()`

`successes` is length 10,000 (1=success, 0=failure). One line: which replicate numbers had success?

H4. Modify `if` for exactly one red

Change the condition so the loop estimates $P(\text{exactly one red})$ — (R,W) or (W,R) — using `!`.

H5. `sample()` with `prob`

Simulate 10,000 people: 70% like eggnog, 30% do not. Use `prob` in `sample()`; store in `eggnog`.

H6. Simulate one room of birthdays

Scenario: Estimate $P(\text{at least two people share a birthday})$ in a room of **23** people (ignore leap years; 365 equally likely birthdays).

Fill in the blanks:

```
n_sims <- 10000
n_people <- 23
match <- numeric(n_sims)
for (i in 1:n_sims) {
  bdays <- sample(1:_____, size = n_people, replace = _____)
  match[i] <- as.numeric(length(unique(bdays)) < n_people)
}
```

What does `match[i] == 1` mean?

H7. Estimate birthday probability

Scenario: Using vector `match` from the birthday loop above (1 = shared birthday in that room, 0 = all different), write **one line** to estimate $P(\text{at least one match})$ among 23 people.

H8. Change room size

Scenario: Same birthday simulation code, but you set `n_people <- 10` instead of 23.

Will `mean(match)` be **larger** or **smaller** than for 23 people? One sentence why.

Answer key — Section H — Birthday matching simulation (demo / exercise)

1. `balls <- rep(c("R", "W"), c(5, 2))`
 2. Loop or replicate with `sample(balls, 2, replace=FALSE)` and indicator; `mean(successes)` $1/7$.
 3. `which(successes == 1)`
 4. Replace if with e.g. `if (sum(draw == "R") == 1) or (draw[1]=="R" & draw[2]=="W") | (draw[1]=="W" & draw[2]=="R")`.
 5. `eggnog <- sample(c("yes", "no"), 10000, replace=TRUE, prob=c(0.7, 0.3))`
 6. Blanks: 365, `TRUE`. `match[i] == 1` means **at least one shared birthday** in that simulated room.
 7. `mean(match)` or `sum(match) / length(match)`
 8. **Smaller** — fewer people means fewer chances for a shared birthday.
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Section I — Assumptions and interpretation (new)

I1. p-value meaning

Which statement is **incorrect**?

- a) The p-value assumes H_0 is true when we compute it.
- b) A small p-value means the observed data would be rare if H_0 were true.
- c) The p-value is the probability that H_0 is true.
- d) We shade tail(s) matching the direction of H_a .

I2. Which BRP condition fails?

You flip the same coin 25 times but the coin is **bent more after each flip** (success probability changes). Which BRP condition fails?

I3. Choosing `pbinom` for $H_a : p > 0.2$

Observed $x = 8$, $n = 25$, $p = 0.2$ under H_0 , $H_a : p > 0.2$. Which line gives the p-value?

- a) `pbinom(8, 25, 0.2)`
- b) `pbinom(7, 25, 0.2, lower.tail=FALSE)`
- c) `pbinom(8, 25, 0.2, lower.tail=FALSE)`

I4. Purpose of `geom_vline`

On a null binomial bar plot, what does `geom_vline(xintercept = x_obs)` show?

I5. `dbinom` vs `pbinom`

When plotting the null distribution with `geom_col`, which function gives bar **heights** $P(X = x)$ for each x ?

I6. Conclusion language

p-value is 0.03 for $H_a : p > 0.10$. Write one sentence conclusion **in context** (not “accept/reject” jargon alone).

Answer key — Section I — Assumptions and interpretation (new)

1. (c) is incorrect — p-value is about **data** extremeness, not $P(H_0 \text{ true})$.
 2. **Fixed** p — success probability is not the same on every trial.
 3. (b) — $P(X \geq 8) = P(X > 7) = \text{pbinom}(7, 25, 0.2, \text{lower.tail}=\text{FALSE})$.
 4. A vertical line at your **observed count** x so you can compare data to the null distribution.
 5. **dbinom** — one probability per bar; **pbinom** gives cumulative probabilities.
 6. Example: If the true proportion were 10%, data this extreme would occur only about 3% of the time — evidence the proportion is **greater than** 10%.
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Section L — Read a pbinom call (identify the test)

L1. ESP right-tail line

Given R line (do not run):

```
pbinom(6, size = 25, prob = 0.2, lower.tail = FALSE)
```

Scenario context: Online ESP test, **5 symbols**, **25 trials**, claim the subject beats random guessing.

1. State H_0 and H_a in symbols.
2. What is n ? null p ?
3. What observed count x is this p-value for? (right tail)
4. Write the p-value as $P(X \dots)$.

L2. Defect left-tail line

Given R line:

```
pbinom(3, size = 25, prob = 0.15)
```

Scenario: Quality team asks whether the defect rate is **below 15%**. **25 items** inspected; **3** defectives observed.

1. H_0 and H_a ?
2. Does this line compute the p-value for this study? **Yes or no** — explain in one phrase.

L3. Gym use right tail

Given R line:

```
pbinom(8, size = 30, prob = 0.20, lower.tail = FALSE)
```

Scenario: Survey **30 students**; research Q: **more than 20%** use the gym weekly; observed **9** yes.

What probability does this compute? Is the observed x for this p-value **8**, **9**, or **10**?

L4. Coin two-sided (conceptual)

Scenario: **20 flips**, observed **16** heads. Test $H_0 : p = 0.5$ vs $H_a : p \neq 0.5$.

A student runs `pbinom(15, 20, 0.5, lower.tail = FALSE)` only.

What tail does this partial line cover? What is still missing for a two-sided p-value?

Answer key — Section L — Read a pbinom call (identify the test)

1. $H_0 : p = 0.2$ vs $H_a : p > 0.2$; $n = 25$, null $p = 0.2$; for observed $x \geq 7$; p-value = $P(X \geq 7)$.
 2. $H_0 : p = 0.15$ vs $H_a : p < 0.15$; **Yes** — $P(X \leq 3)$ is the left-tail p-value for $x = 3$.
 3. $P(X \geq 9)$; observed $x = 9$ (upper tail uses `pbinom(x-1, ..., lower.tail=FALSE)`).
 4. Right tail $P(X \geq 16)$; still need the **left tail** equally extreme counts for two-sided p-value.
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Section M — P-value from story + given pbinom line

M1. Left-handed school

Scenario: 25 students surveyed; national left-handed rate 10%. Research Q: is the school rate **higher**? Observed 6 left-handed.

A classmate runs:

```
pbinom(5, size = 25, prob = 0.10, lower.tail = FALSE)
```

1. What probability does this compute?
2. Is this the correct p-value for this study? **Yes or no.**

M2. Defect rate — wrong tail

Scenario: 25 items, 3 defectives. Test whether rate is **below 15%**: $H_0 : p = 0.15$ vs $H_a : p < 0.15$.

Someone runs `pbinom(3, 25, 0.15, lower.tail = FALSE)`.

1. What probability does that line compute?
2. Is it the correct p-value? If not, give the correct one-line setup.

M3. ESP — interpret output

Scenario: ESP test, 25 trials, $H_0 : p = 0.2$, $H_a : p > 0.2$, observed 8 correct.

`pbinom(7, 25, 0.2, lower.tail = FALSE)` returns **0.035**.

Write a one-sentence conclusion about ESP **assuming** H_0 is true (use the p-value, not “accept/reject” jargon alone).

Answer key — Section M — P-value from story + given pbinom line

1. $P(X \geq 6)$ under $H_0 : p = 0.10$; **Yes** — right tail for $H_a : p > 0.10$ at $x = 6$.
 2. Computes $P(X \geq 4)$ (wrong tail). Correct: `pbinom(3, 25, 0.15)` for $P(X \leq 3)$.
 3. If guessing only, data this extreme (8+ correct) would occur only about 3.5% of the time — unusual under H_0 .
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